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A Speech Driven Framework For Identifying Low Resourced Ethnic Languages of Bangladesh

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***Abstract*—The ethnic languages with limited resources are not well represented in the field of computational speech, which makes it challenging to implement into the new technology. This study introduces a speech-based machine-learning system to detect 3 low-resource ethnic languages of Bangladesh that are Chakma, Marma and Garo. A collection of 2,321 audio samples of different native speakers was gathered, reading translated versions of 211 sentences in Bangali. The samples were checked to be linguistically correct and they were normalized in WAV format. Noise, silence, normalization and Mel Frequency Cepstral Coefficient (MFCC) extraction were performed as the preprocessing steps to obtain the consistent feature representations that can be used in machine-learning analysis. Through MFCC features, a set of classical machine-learning models, Support Vector Machine (SVM), Logistic Regression, Random Forest and K-Nearest Neighbors, were trained. The accuracy of the SVM and Random Forest models were the highest, with both models obtaining accuracy of 99.35%, whereas the accuracy of the Logistic Regression and KNN is 99.14% and 98.92% respectively. These good findings indicate that classical machine-learning approaches can be used to reasonably well learn the acoustic patterns of the differences between Chakma, Garo, and Marma, even with a relatively small dataset. In order to improve interpretability, SHAP (SHapley Additive exPlanations) was used on the SVM and the Random Forest models. SHAP outcome indicated that the most valuable features in all models were mfcc5mean, and the MFCCmean category was the most significant. On the whole, the results support the possibility of constructing an effective and explainable system of language-identification of low-resourced Bangladeshi languages with the help of classical machine learning. The future research consists of dataset expansion, deep-learning implementation, as well as real-time deployment to further digital inclusion and language preservation.**

***Index Terms*—Language Informatics, Ethnic Language, Natural Language Processing, Ethnic Language Identification, Multilingual Speech Processing.**

I. Introduction

In Bangladesh, there are a number of indigenous communities, whose languages, such as Chakma, Marma and Garo, are endangered and threatened critically as there is less intergenerational passing on of the language, less literacy among people, and no digital materials of learning. The sociolinguistic research shows that there is a tendency to decrease the use of language among the Chakma speakers, and the same problems are reported among the Garo community. Indigenous scripts are also examined historically and structurally which further portrays the absence of full linguistic documentation of these languages. This cultural vulnerability and technological negligence create an acute need to use computational tools in helping to preserve, make accessible, and revitalize such artifacts. Modern works are nowadays filling up the gaps of these low resourced works. In the case of Chakma, masked language modeling using transliteration has proven the capabilities of neural systems even in severely limited environments [1] and ChakmaNMT has shown that machine translation is possible even with limited resources [2]. The MELD corpus increased the length of text in Chakma, Marma and Garo by means of multilingual corpora based on Bengali script [3]. In spite of these developments, the literature is all textual and therefore, the speech modality which is the major mode of communication in the indigenous societies is not well explored. In the meantime, the applicability of acoustic features and statistical models in the area of low-resource spoken language identification has been proven in world research. The research on the Ethiopian languages [4] and the Tamil dialects [5] proves that MFCC, spectral cues and prosodic dynamics are sufficient to distinguish between related language systems. Other audio classification studies such as music genre classification and neural audio modeling also demonstrate that spectral and rhythmic patterns are highly discriminative in brief and noisy audio samples. Such results offer a powerful methodological basis to the development of speech-based instruments of indigenous languages of Bangladesh. One of the key changes that allow this kind of research has been the recent publication of the first multi-ethnic speech sample of Chakma, Marma, and Garo, which consists of 2,321 utterances of native speakers in various contexts [6]. Nevertheless, there is no existing literature that has tried to identify speech based, characterize acoustically or model comparatively these languages with classical architecture, neural architecture or self-supervised architecture. As a solution to this gap, the current paper presents a speech-based paradigm of recognizing low-resource ethnic languages in Bangladesh, incorporating classical machine learning models (Random Forest, SVM, Logistic Regression, KNN).

In addition to the classification, Explainable AI (XAI) is included in the study to understand the acoustic formations of model decisions. Based on these concepts, the current study

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uses SHAP which MFCC coefficients, chroma features and time-frequency structures help to most distinguishably differentiate Chakma, Marma and Garo speech. This method with the XAI enhances transparency, facilitates the identification of bias in a dataset (e.g. speaker or device confounds), as well as provides linguistic information in the form of acoustic patterns distinguishing these endangered languages. On the whole, the presented research makes the initial computational speech model of Chakma, Marma and Garo, an in-depth assessment of traditional and state-of-the-art models in low-resource settings and an XAI-based explanation that contributes to the scientific knowledge and responsible creation of local speech technologies.

II. Literature Review

The study of low-resource and endangered languages has been a continually increasing area of research over the last ten years, as NLP and speech technologies have been necessary to be extended to marginalized groups of linguistic communities. In the case of the native languages of Bangladesh Chakma, Marma and Garo, most of the computational research is conducted in text processing but not speech, so there are still many gaps in acoustic modeling and spoken language recognition.

Khisa et al. suggested a masked language modeling (MLM) paradigm based on transliteration to resolve script scarcity in the Chakma language that is critically low-resource. They found in their work that cross-lingual knowledge transfer using transliteration can be very effective in downstream NLP performance in resource-constrained environments. The study however only targets textual modeling but not speech or acoustic representations. [1]

Chakma et al. made the first neural machine translation system of the Chakma language called ChakmaNMT. The model produced encouraging translation quality by utilizing script normalization strategy and low-resource adaptation strategy. Although it contributed to textual NLP, the work leaves out the processing of spoken language and identification of speech-based language. [2]

Mahi et al. introduced MELD, a multilingual ethnic text corpus, in Bengali script with English and Bengali translation of Chakma, Marma and Garo. The dataset does not support speech data and is limited to acoustic modeling and speech technologies as it facilitates cross-lingual and comparative text-based research. [3]

Wondimu and Tekeba suggested a MFCC-based system based on Gaussian Mixture Model (GMM) in identifying Ethiopian languages. They found that using low-resource conditions, acoustically similar and related languages can be successfully differentiated, which defines a methodological basis of spoken language identification. [4]

Nanmalar et al. attempted to identify literature and colloquial dialect of Tamil based on acoustic and prosodic characteristics. Their experiment established that MFCC, spectral and prosodic cues can encode phonotactic and rhythmic differences and thus they are effective in differentiating speech varieties which are closely related. [5]

Anik et al. published the first publicly available dataset of Chakma, Marma and Garo speech, which comprised 2,321 utterances of the native speakers. Even though the dataset not only addresses an essential resource gap in the study of indigenous speech in Bangladesh but also provides a wide range of speech modeling and language identification experiments, it lacks baseline speech modeling and language identification experiments. [6]

Liang et al. used spectral and timbral features of the Million Song Dataset to perform music genre classification at a very large scale. They found that MFCC and spectral representations are very discriminating ones in terms of their use in audio classification and acoustic features prove to be effective in non-linguistic audio modeling. [7]

Mamun et al. used the neural network models to analyze Bangla music genre based on MFCC and spectral features. Their findings showed that it is possible to have a high-performance with acoustic feature-based neural models in low-resource audio settings, which supports the appropriateness of acoustic features to speech-related classification problems. [8]

Brata et al. built an automatic scoring system of pronunciation on mobile-assisted learning through the use of pitch, volume, and rhythm. Their results emphasize the role of prosodic and temporal elements of speech evaluation but the research does not refer to language identification or low-resource ethnic languages. [9]

Sarno et al. suggested a music fingerprinting methodology based on Bhattacharyya distance on song and cover-song recognition. The experiment proved that spectral similarity distances are very robust to detect acoustic patterns and can be used to give knowledge which is transferable to speech-based discrimination tasks. [10]

Bahuleyan offered a detailed paper on the classification of music genre with the help of machine learning methods, with a focus on the MFCC-based representations and classical classifiers. The effectiveness of lightweight acoustic models is supported by the work and is especially appropriate in low-resource and real-time applications. [11]

Wu et al. formed the concept of Usable Explainable AI (XAI), a strategy of ten approaches in constructing the transparent, human-centered and reliable AI systems. They have made the aspect of explainability vital to trusting, being fair and deploying in the real world more especially in socially sensitive applications like language technologies. [12]

The article by Cambria et al. addressed the importance of explainability in the context of modern language models using the concept that XAI strengthens, enhances interpretability, and ethical AI implementation. Their results support the idea of including the explanation mechanisms in the speech and language systems, particularly in the case of underrepresented communities. [13]

III. Methodology

We construct language identification system of speech based on low-resource corpus of Chakma, Marma and Garo, consisting of 2,321 samples, acquired in a controlled recording situation and comprised of native speakers. In order to guarantee consistency and reproducibility, the data are standardized through annotation, metadata organization, normalization, removal of noise, removal of silence and segmentation. The fundamental feature space that is to be employed in modeling the delay, is the extraction of acoustic representations on the basis of MFCCs spectral and delta as well as delta-delta coefficient and prosodic representations. Classical machine learning algorithms (SVM, Random Forest, KNN, Logistic Regression) are trained and their efficacy is evaluated with the help of traditional measures of accuracy, precision, recall and F1-score. Classical machine-learning models have been chosen because they provide stable and easily interpret-able results in the case of extreme low-resource conditions, unlike deep and self-supervised models, which often need larger and more varied datasets to prevent overfitting. The theory behind

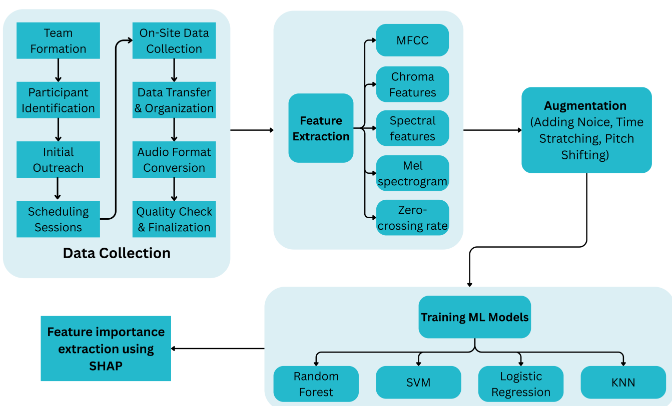

Fig. 1: Full Workflow

the methodology is the low-resource speech processing theory and its implementation is through Explainable AI (SHAP) to understand the model selection and to understand the discriminative acoustics features across languages. The above Figure 1 represents the entire methodological approach.

A. Data Collection

The speech dataset has been created as a result of a well-organized and community-based data collection procedure in order to guarantee authenticity and quality. There were eleven native speakers aged 20-26 and volunteered to take part, 5 Chakma (3 males, 2 females), 3 Marma and 3 Garo speakers. They were called upon using the existing university ethnic networks to ensure authentic linguistic representation. The subjects were given 211 Bengali sentences with a predetermined list through Google sheets, which they translated and read out in their native language through transliteration in Bengali script. Individual recordings were gathered in controlled sessions in which there were 2,321 audio clips that lasted between 1 and 7 seconds. Language labels and contextual frequencies were annotated on all the samples and cross-speaker validation was done by recording the same sentences across speakers of the same ethnicity. The ultimate validation was done in conjunction with university student ethnic community clubs to ensure that there was authenticity of the speakers and consistency in pronunciation so that they are able to have a sound corpus to use in identifying the language through speech.

B. Data Pre-Processing

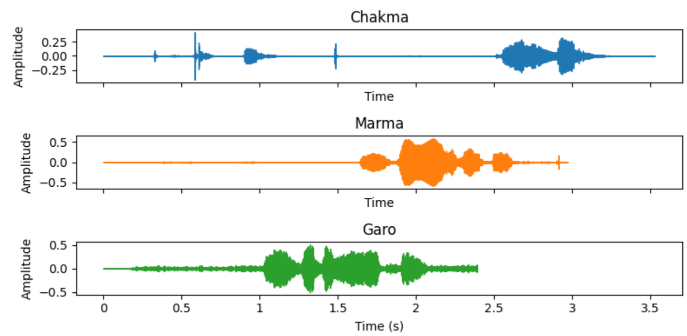

Fig. 2: Wavelength of Chakma, Marma & Garo Language

The Figure 2 visually shows us how the raw data looks like in audio frequency. The raw records will undergo the standard pre-processing procedures, which, in addition, are required in order to ensure uniformity of the data set. It achieves this through noise elimination, silence clipping, amplitude correction and segmentation. The preprocessing will be aimed at eliminating the environmental conditions caused by the actual capture of audio in the real world and also enabling the speakers to have even conditions. Afterwards, all the audio files that have been processed are prepared to be feature extracted. Formation of MFCCs is the key property of this study. The MFCC process is likened to the human auditory system because it converts spectral energies to the perceptually-motivated mel-scale. Outputs indicates three important encapsulations of a Chakma speech signal. The upper waveform shows the raw audio amplitude versus time, which can be seen to identify the periods of speech and the periods of silence. The audio is converted to a time-frequency energy in the Mel-spectrogram in the middle. Mel-spectrogram emphasizes dominant frequency components in the speech segments. Lastly, the MFCC plot below picks up on compact, speech specific acoustic features that are often utilized in machine learning models. Combining these visualizations will give an in-depth perspective of the Chakma audio behavior in terms of time, frequency and feature space.

The steps of calculating the MFCC are:

1) **Framing and Windowing:** The audio signal is divided into short overlapping frames (typically 2048 samples with a hop size of 512 samples). A Hann window is applied to each frame to reduce edge discontinuities.

2) **Fourier Transform:** The Short-Time Fourier Transform (STFT) is applied to each framed signal to obtain its frequency spectrum.

$$X(k,n) = \sum_{m=0}^{M-1} x[n+m]\, w[m]\, e^{-j\frac{2\pi km}{M}} \tag{1}$$

3) **Mel Filterbank Processing:** The magnitude spectrum is passed through a set of triangular filters spaced according to the mel frequency scale.

$$E_m = \sum_{k=k_{\text{low}}}^{k_{\text{high}}} |X(k)|^2 H_m(k) \tag{2}$$

4) **Logarithmic Compression:** The energies of the mel filterbank outputs are converted to the logarithmic scale.
5) **Discrete Cosine Transform (DCT):** The DCT is applied to decorrelate the filterbank energies, resulting in compact MFCC coefficients.

$$c_i = \sum_{m=1}^{M} L_m \cos\left(\frac{\pi i(m-0.5)}{M}\right), \quad i = 1, 2, \ldots, 13 \tag{3}$$

Since 13 coefficients were obtained in each frame, the mean interframe sum was employed to give a 13 dimensional feature vector of each audio file.

C. Data Description

The dataset is publicly accessible in the form of digital repository [6], and it includes 2,321 short speech records in three ethnic languages of Bangladesh including Chakma, Marma, and Garo. A recording of a single 1-7 seconds utterance, either of 11 native speakers (Chakma: 3 male, 2 female; Marma: 3 male; Garo: 3 male), is recorded, and 211 sentences are used per speaker. The recordings were made of raw voice information of every speaker.Then normalization and translation were done during the preprocessing. The data is sorted into language folders, and each speaker and file have subfolders, and the name of the files is the language, speaker, and sentence index making it easy to reuse and access the data in speech recognition, language identification, and low-resource NLP studies.The structure of the dataset is illustrated in Table I.

TABLE I: Dataset Description

Dataset	Speakers	Sentences	Min (sec)	Max (sec)	Avg (sec)
Chakma	5	1055	1.416	7.019	2.923
Marma	3	633	1.681	5.377	2.872
Garo	3	633	1.460	4.277	2.349
Total	**11**	**2321**	–	–	–

Table II contrasts this proposed audio dataset with existing text-based resources regarding their different properties and indicates that our dataset is the only one providing recordings.

TABLE II: Comparison of Existing Datasets and Our Proposed Dataset

Dataset	Class	Linguistics	Type	Dataset Size	Sentence	Audio
ChakmaNMT (2024) [2]	3	Chakma, Bangla, English	Text	15,021 parallel 42,783 monolingual	✓	×
MELD (Mahi et al., 2025) [3]	3	Chakma, Garo, Marma	Text	3,046 sentences	✓	×
Ashaduzzaman and Rashel	1	Marma (only)	Text	Not mentioned	✓	×
Our Dataset [6]	3	Chakma, Garo, Marma	Audio	2,321 recordings	✓	✓

D. Analysis

The approach taken in the given study is quite appropriate to the limitations of low-resource speech processing and well-balanced in terms of performance, interpretability, and computational efficiency. MFCC-based acoustic feature including their coefficients of delta and delta-delta were also important since they not only record the spectral envelope of speech but also the temporal variations of the speech that have been known to reflect language-specific phonetics. Further spectral (centroid, bandwidth, roll-off) and prosodic parameters were added to the representation, which made it rich enough to be discriminated even with small amounts of training data. Strong preprocessing procedures such as normalization, noise elimination, silence cutting, and segmentation played an important role in minimizing acoustic variability due to recordings and speakers, thus enhancing the generalization of the model. Classical machine learning algorithms like SVM and Random Forest were carefully selected instead of the heavy deep models to make sure that they are reliable in the conditions of small amounts of data but retain high accuracy. Lastly, the addition of SHAP-based explainability to the methodological framework reinforced the framework that model decisions were based on the linguistically meaningful acoustic features as opposed to artifacts induced by chance.

IV. Results and Discussions

A. Model Performance

The speech-based ethnic language identification system is experimentally tested and proves that under the conditions of low resources, classical machine-learning models are capable of high reliability performance. Table III shows us, of the four classifiers, namely K-Nearest Neighbors (KNN), Logistic Regression, Random Forest and the Support Vector Machine (SVM) classifier, the most reliable and consistent model when it comes to the division of Chakma, Marma and Garo speech samples is the Random Forest and the Support Vector Machine (SVM) classifier because they both gave the highest overall accuracy of 99.35

TABLE III: Comparison of Accuracy for Models

Models	Accuracy	Precision	Recall	F1 Score
Random Forest	0.994	0.992	0.992	0.992
SVM	0.994	0.993	0.993	0.993
Logistic Regression	0.991	0.990	0.992	0.991
KNN	0.989	0.988	0.988	0.988

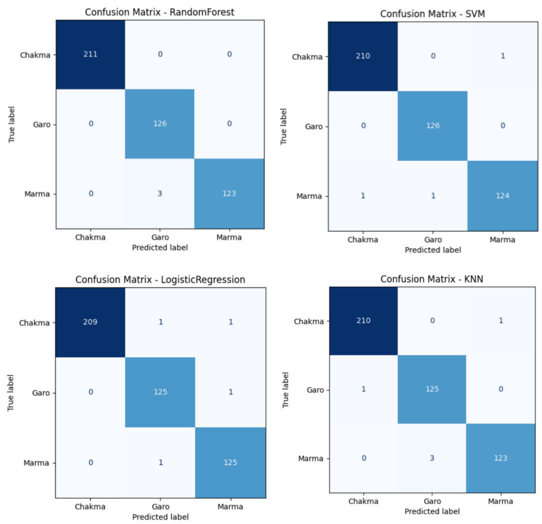

Fig. 3: Confusion Matrix of Using Models

The Figure 3 contains the confusion Matrix for gained models. Their accuracy, recall and F1-scores were also almost similar among all classes and there was great generalization in spite of the different language patterns and little training data. KNN classifier, though slightly lower with the accuracy of 98.92% was very accurate with Chakma and Marma but a bit confused with Marma and Garo. From table-III we see SVM and RandomForest performing the best. We did some SHAP analysis on SVM. The SHAP (SHapley Additive exPlanations) analysis is the one that measures or quantifies the contribution of each group of features to the decision making process that the SVM classifier takes. Although the model accuracy results show the performance of the model, SHAP will describe the reason behind the performance of the model by showing the relative importance of various categories of acoustic features. Figure 4 displays the accuracies achieved from the models using a bar chart.

B. Model Explainability

From above Figure 4 we can observe how different features play different roles for the model to detect accurately. This can be explained by the fact that, as the explainability analysis shows, MFCC-based features are significantly predominant in the process of decision making of the ethnic language identification system that is speech-based. Of all the extracted features, MFCCmean is the most influential (0.0090), then MFCC standard deviation (MFCCstd) and MFCC delta coefficients. This high dependency on MFCC-related descriptors shows that spectral envelope properties of speech, representing

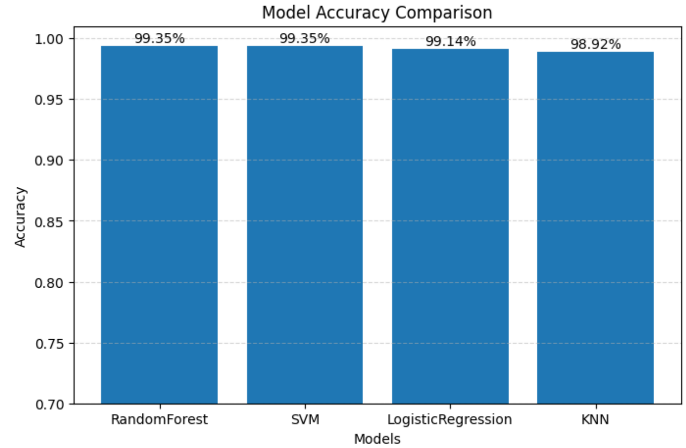

Fig. 4: Bar Chart of ML Model Accuracies

phonetic structure, formant distribution and the overall timbral patterns, are very discriminative in differentiating between Chakma, Marma and Garo.

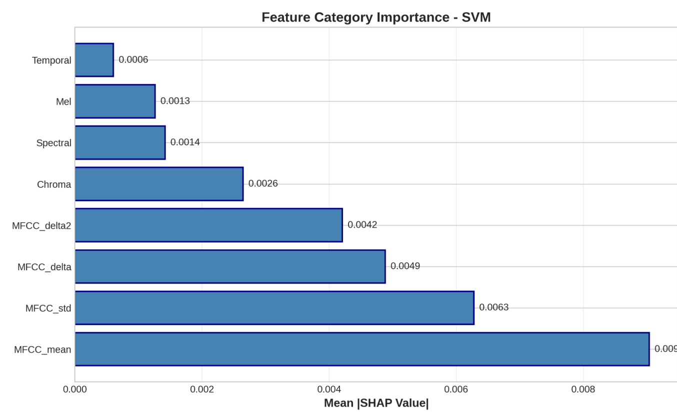

Fig. 5: Feature Importance Using SHAP Plot

The nature of these properties is that they are inherently different across languages since there are differences in articulation, phonotactics, and vocal tract structure and thus MFCCs are especially useful in identifying audio recorded language.The dynamic speech properties also add to the performance in the classifications. The results of the high influence of MFCCdelta and MFCCdelta2 show the importance of the temporal variation in articulation and prosodic transitions. The dynamic characteristics allow the SVM model to reconcile the changes in spectral properties over the time, which is the key to separating languages that share similar fixed acoustic patterns. Conversely, Chroma (0.0026) and spectral characteristics (0.0014) have a moderate contribution meaning that pitch-related harmonic information and the general distribution of spectral energy contribute to language discrimination in a secondary but facilitating way(mentioned in Figure 5). Lastly, Mel-band energy features (0.0013) and features of time (0.0006) have the lowest impact on the decisions of the model. The comparatively low SHAP values are an indication that coarse temporal statistics, i.e. energy and zero-crossing rate, and wide Mel-scale energy representations contain relatively little discriminatory information with respect to this particular task. This indicates that fine -grain spectral and dynamic cues instead of global time trends play a larger role in recognizing similar low-resource ethnic languages.

V. Discussion

The experimental findings indicate that the classical machine-learning models are very effective in the speech-based recognition of the low-resource ethnic languages at the cost of well thought-out acoustic features and preprocessing. Random Forest and SVM obtained the largest general accuracy (99.35%), which suggests that the two methods are better placed to model non-linear decision boundaries and at least partially deal with overlapping acoustic features, which is also in line with the results in the previous low-resource studies on spoken language identification. Although the performance of the classification is high, this study works with a set of low-resource requirements. The data consists of the records of a small number of speakers, and increasing imbalance across languages and gender is inevitable as far as it is dependent on the availability of volunteers. Since several utterances are produced by the same speakers, it is possible that the models to some extent represent speaker-specific acoustic characteristics along with language-specific tendencies. Although this effect can be minimized by constant preprocessing and cross-speaker recording sentence, the findings can be viewed as preliminary. The next steps in work include increased diversity of speakers, balanced demographics and fully speaker-independent evaluations to enhance generalization. SHAP-based explainability showed that the most significant features were MFCC mean, variance, and temporal derivatives, which confirms that spectral envelope and dynamic articulation cues play an important role in making the distinction between these languages. Although the performance is high, there are still drawbacks because of the small and scripted dataset and the possibility of speech overlapping because of the speakers which could lead to generalization to spontaneous speech and speakers that are not known. Future directions are to increase the size of the corpus in terms of number of speakers and environments, specify more sophisticated forms of augmentation or self-supervised representations, and refine explainability further in a way that is able to separate linguistic characteristics and speaker-specific artifacts.

VI. Conclusion

The study has introduced a speech-based work that is easy to implement and understand and identify three low-resource ethnic languages in Bangladesh, Chakma, Marma and Garo. Through the sophisticated speech data curation, powerful preprocessing, MFCC-based acoustic features, and state-of-the-art machine learning algorithms, the suggested framework demonstrated great classification, Random Forest and SVM attained 99.35%. The findings show that resource-efficient models combined with effective feature engineering can effectively solve language identifying tasks even in the conditions of extreme data limitations. The Explainable AI integration using SHAP contributed to the transparency further as it demonstrated the contribution of acoustic features to the model decisions supporting the trust, fairness, and linguistic interpretability. Although drawbacks of datasets size, scripted speech and speaker dependency have not been eliminated, the work has provided a solid foundation on future speech technologies of indigenous languages. This study will provide a viable, ethical, and scalable framework on which to develop low-resource speech processing and help keep endangered language communities sustainable in Bangladesh in the long run.

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